Python Problems

Problem 1: The Fairest Grade (Lists)

Mr. Rahman is grading a quiz, but he has a rule: he always drops the single highest score before calculating the class average, so one lucky genius doesn't skew the curve.
Given a list of scores, write code that:
1. Finds the maximum score in the list.
2. Removes one occurrence of that maximum score.
3. Calculates and prints the average of the remaining scores.

```
scores = [78, 92, 85, 92, 60, 74]
# Expected: remove one 92, then average the rest
```

Problem 2: The Grocery Ledger (Dictionaries)

Fatima runs a small grocery stall. She keeps a dictionary where the keys are item names and the values are their prices.
Write code that:
1. Stores at least 5 items and their prices in a dictionary.
2. Prints the name and price of the most expensive item.
3. Prints the total cost if someone buys everything on the list once.

```
prices = {
    "rice": 60,
    "milk": 45,
    "eggs": 12,
    "bread": 35,
    "oil": 150
}
```

Problem 3: The Attendance Counter (Functions)

Every morning, a teacher calls out student names, and marks who is present in a list. Write a function called count_present that takes a list of names (with possible repeated names if someone got marked twice by mistake) and returns how many unique students were present.

```
def count_present(names):
    # your code here
    pass

attendance = ["Amina", "Karim", "Amina", "Nabil", "Karim", "Sara"]
print(count_present(attendance))  # should print 4
```

Problem 4: The Temperature Reporter (Functions + Lists)

A small weather station records the temperature (in Celsius) every hour of the day. Write a function hottest_and_coldest that takes a list of temperatures and returns a tuple with the highest and lowest temperature recorded.

```
def hottest_and_coldest(temps):
    # your code here
    pass

readings = [30, 32, 29, 35, 31, 28]
print(hottest_and_coldest(readings))  # should print (35, 28)
```

Problem 5: The Library Card (Classes)

Create a simple class called Book that represents a book in a small library.
Requirements:
1. The class should store title, author, and is_borrowed (a boolean, default False).
2. Add a method borrow() that sets is_borrowed to True and prints a message.
3. Add a method return_book() that sets is_borrowed to False and prints a message.

```
class Book:
    def __init__(self, title, author):
        # your code here
        pass

    def borrow(self):
        # your code here
        pass

    def return_book(self):
        # your code here
        pass

my_book = Book("The Old Man and the Sea", "Ernest Hemingway")
my_book.borrow()
my_book.return_book()
```

Problem 6: The Fair Average (Lists, revisited)

A judge is scoring a talent show with 5 judges. To avoid bias, the show organizers drop both the highest and the lowest score before averaging the rest.
Given a list of 5 scores, write code that:
1. Removes one occurrence of the maximum score.
2. Removes one occurrence of the minimum score.
3. Prints the average of the remaining 3 scores.

```
judge_scores = [8, 9, 7, 10, 6]
```

Submission Instructions

Implement all the problems above sequentially in a single Google Colab notebook, **making sure the file access is set to “Anyone with the link can view.”**

Then submit a .txt file named in the format **<id>_<name>.txt** (for example: **2021304_MirajHossain.txt**) containing only the Colab link, e.g.:

```
https://colab.research.google.com/drive/ua09sdu93qej3o
```